FoliumScope

Presented by TEAM ERROR404

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The Team

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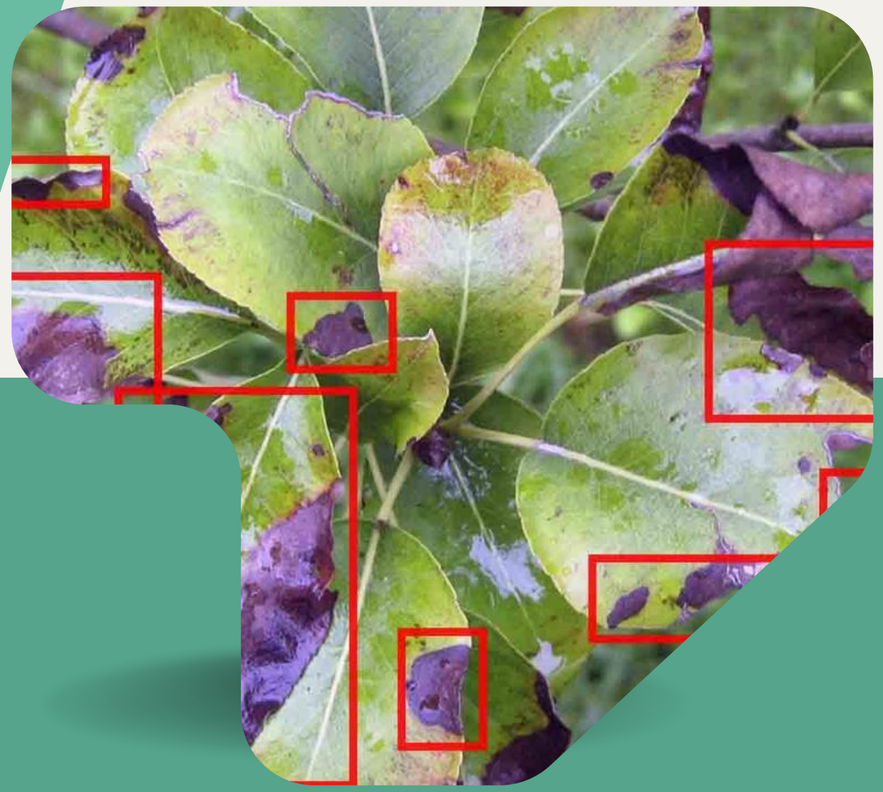
What is the FoliumScope ?

"FoliumScope" is our name for a machine learning model that uses advanced technology to closely analyze the health of plant leaves. This name is a combination of "folium", which is Latin for leaf, and "scope", which implies the ability to examine or observe closely.



Our Track

Plant disease detection

Develop a machine learning model that can accurately identify and segment regions of plant leaves affected by disease/pest in images provided in a dataset. The model should be able to classify the images into different disease categories and highlight the affected areas on the leaves. The participants will be judged based on the accuracy, inference time and novelty of the architecture used.

The Solution

To improve the novelty of the architecture, we can use techniques such as tensorflow, keras, transfer learning, where we use a pre-trained CNN model as a starting point and fine-tune it on our specific task. We can also use attention mechanisms, which can help the model to focus on the most relevant parts of the image when making predictions.

Overall, the success of the model will depend on the quality and diversity of the dataset, the accuracy of the annotations, and the effectiveness of the designed architecture.

TECHSTACKS **USED**

Librarie
s

Tensorflo
w

Kera
s

Flask

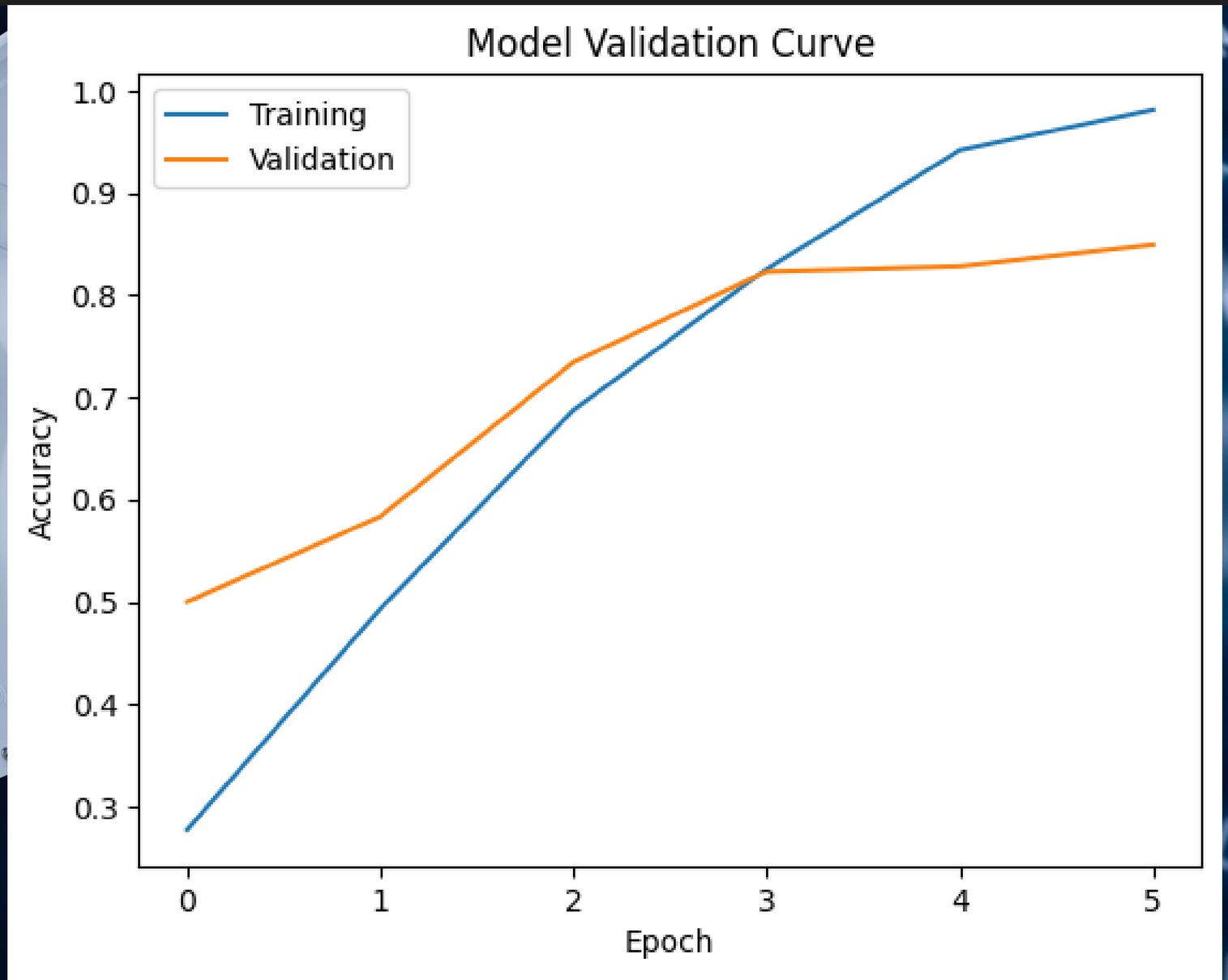
"Flask" *will be used for deployment of Machine Learning Model.*

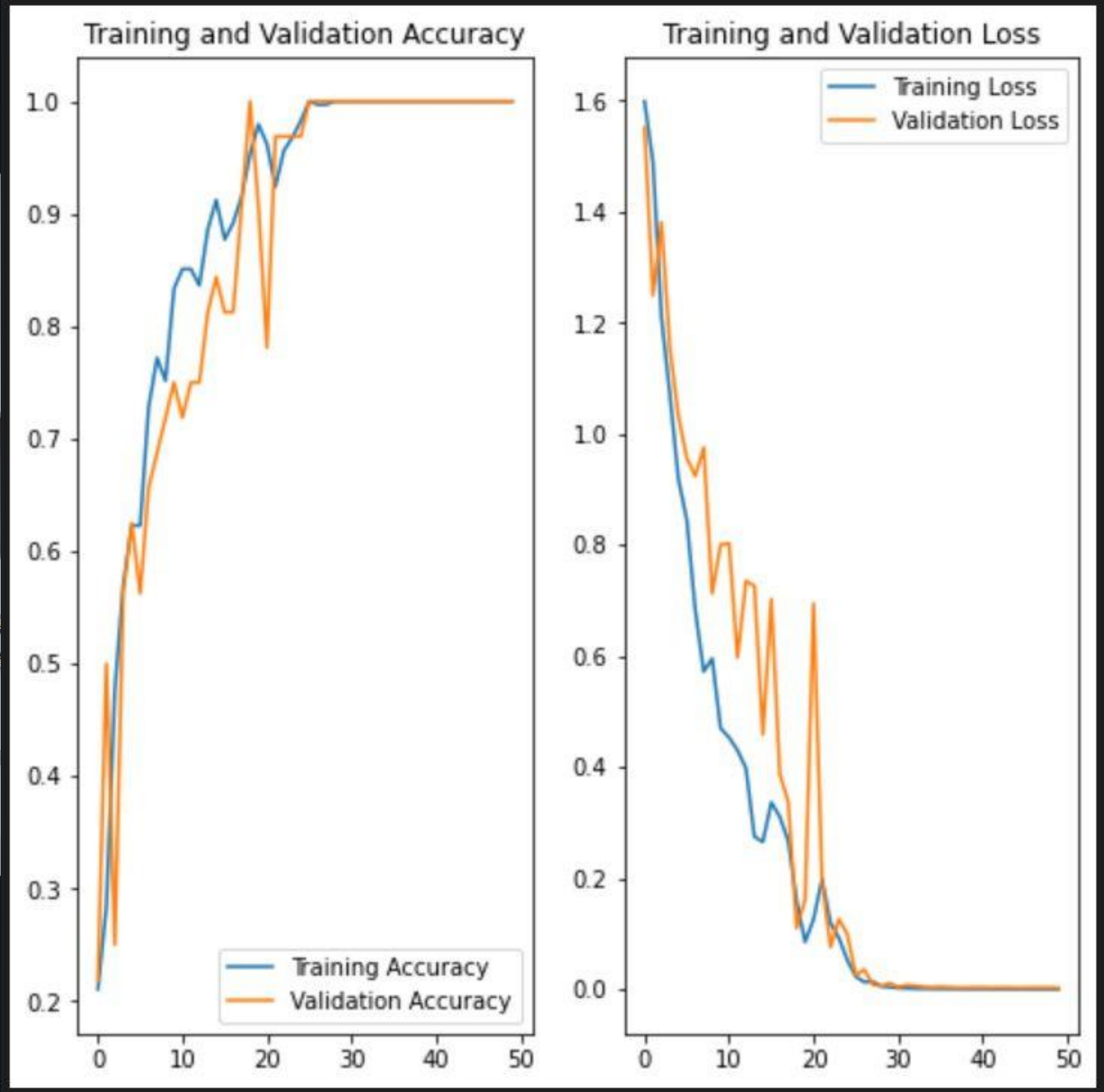
OpenCV

```python
max(val_acc)
```

[32] ✓ 0.0s Python

```
0.8489583134651184
```




Training and Validation Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
10
20
30
40
50
Training Accuracy
Validation Accuracy
Training and Validation Loss
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
50
Training Loss
Validation Loss

FoliumScope

Plant disease detection

Choose .

FoliumScope

Help

Plant disease detection

Choose...

Result: Ring Spot

Thank You

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